

Solution Requirements

Date	15 JULY 2026
Project Name	Smart Lender: AI-Powered Loan Approval Prediction System
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users (Bank Officer/Admin) securely log in to access the loan prediction system.	High
2	Authorization levels	Admin manages the system and ML model, while Credit Officers can submit applicant details and view predictions.	High
3	External interfaces	Integrates the trained XGBoost model with the Flask web application deployed on IBM Cloud.	Medium

4	Transactions processing	Accepts applicant details, validates inputs, predicts loan approval, and displays the result instantly.	High
5	Reporting	Displays loan approval/rejection results and prediction probability for each application.	Medium
6	Business rules, etc.	Loan approval is determined based on applicant information such as income, credit history, loan amount, employment status, and property area using the trained ML model.	High
7	Compliance to laws or regulations	Protect applicant data and ensure secure handling of financial information according to organizational privacy policies.	Medium
8	Other	Allow future retraining or updating of the machine learning model with new datasets.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Generate loan prediction quickly after user submission.	Prediction within 2 seconds

2	Scalability	Support increasing numbers of loan applications and future dataset expansion.	Handle 1000+ applications/day
3	Security & Data Privacy	Protect applicant information through secure storage and authenticated access.	Authorized users only
4	Reliability & Availability	Ensure continuous availability of the prediction system.	99% uptime
5	Usability & Accessibility	Provide a simple, responsive, and easy-to-use interface for users.	Responsive on desktop and mobile
6	Other	Maintainable and modular code for easy updates and model replacement.	Easy deployment and maintenance